42 Circulation and Gas Exchange

KEY CONCEPTS

Study Tip

Draw and annotate a diagram: Sketch a simplified double circulation system like the idealized one on this page. Label where each of the following is highest: blood pressure, CO_2 concentration, O_2 concentration, and total cross-sectional area of the blood vessels. Note that the left side of your diagram represents the right side of the heart in the body, and remember "ReViTaLize" to help remind you of the path of blood from the Right Ventricle to the Lungs.

Go to Mastering Biology

For Students (in the eText and Study Area)

- Get Ready for Chapter 42
- Figure 42.10 Walkthrough: Interrelationship of Blood Vessel Area, Velocity, and Pressure
- Figure 42.22 Walkthrough: Structure and Function of Fish Gills
- BioFlix® Animation: Gas Exchange in the Human Body

For Instructors to Assign (in the Item Library)

- Animation: Gills Are Adapted for Gas Exchange in Aquatic Environments

Ready-to-Go Teaching Module (in Instructor Resources)

- Cardiac Cycle and Heart Function

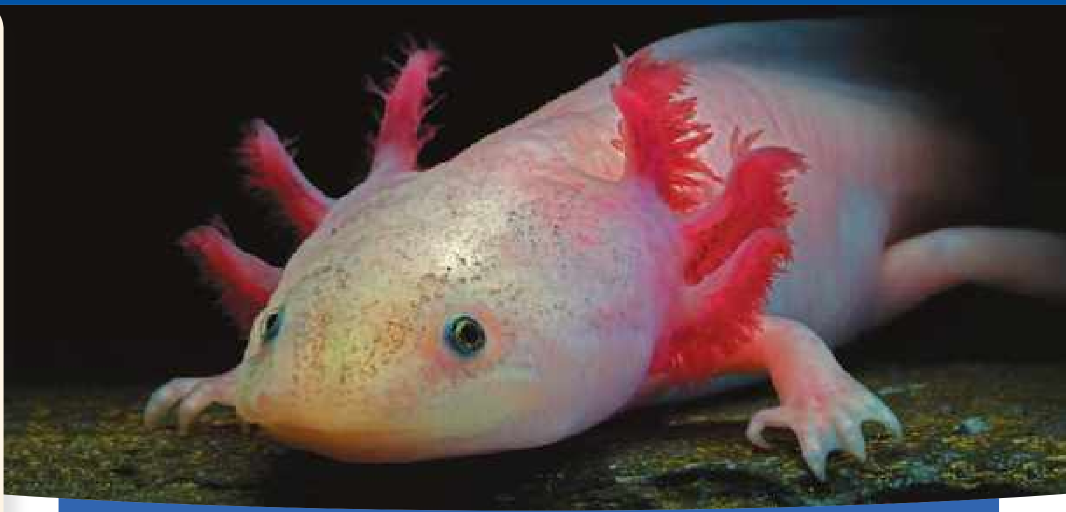

Figure 42.1 This animal may look like a creature from a science fiction film, but it's actually an axolotl, a salamander native to shallow ponds in central Mexico. The feathery red appendages jutting out from its head are gills. External gills, although uncommon in other adult animals, help axolotls carry out a process common to all organisms: the exchange of substances between body cells and the environment.

How are structure and function related in the exchange and circulation of oxygen and carbon dioxide?